

F718 — CHARTER (Insight-tier, prompt-28b, chartered to Lyra): quark generation = radial distance to the Shilov boundary — and that ONE quantity controls both the mass-ORDERING (already DERIVED) and the mass-PRECISION (the new lead). The convergence that opens it: Casey’s observation + the tier data — quark mass-ratio precision degrades monotonically up the generations (m_u/m_d 0.09% $\rightarrow$ m_s/m_d 0.5% $\rightarrow$ m_c/m_s coarse $\rightarrow$ m_t/m_b 1.7% $\rightarrow$ V_{cb} 8%), while the leptons are the control and stay sharp all the way up (p/e 0.002%, m_μ/m_e 0.004%, $m_\tau/m_e \sim 0.05\%$). So the blur tracks confinement/boundary-reach, not generation number. The lens is one already-derived object read a new way — and the whole thing shares its one computation, the discrete-series address (a,b), with the generation count and the y_t/m_t radial overlap: ONE dig, THREE payoffs. Tier: LEAD / INSIGHT — NOT a claim. The convergence (Casey’s intuition + the corpus’s derived ordering) says the lens is *right*, it is NOT evidence the precision-prediction holds. Cal #27 fires HARDEST here precisely because it’s clean — the earning test is quantitative (does boundary-distance *predict* the blur, not merely track it monotone-by-eye).

Lyra, Tue 2026-07-28 09:40 EDT. Chartering the quark-generation lens, corpus-grounded (F603/K768 = the derived ordering half; the confined-quark-no-pole-mass = the softness). The new half (boundary-distance $\rightarrow$ precision-loss) is a lead I must *compute*, not bank. And it’s the same radial-address dig the generation count needs — so I run them together.

The two facets of one object

- Boundary-distance $\rightarrow$ mass ORDERING: already DERIVED (F603/K768). Mass = the fermion mode’s radial overlap with the boundary condensate O (the SO(5)-vector Higgs, spherical, boundary-reaching, F603); “mass ordering = boundary-reach ordering,” $t > c > u$ = most $\rightarrow$ least boundary-localized. The top is the *outermost* (near-boundary) gen-3 mode. Solid, not new.
- Boundary-distance $\rightarrow$ PRECISION-LOSS: NEW, the open lead. Near the boundary, the radial overlap is a limit — “approaches but never reaches” (F603) — not a pinned bulk cell. A limit is intrinsically *soft*: the observable itself is scheme/scale-dependent (a confined quark has *no clean pole mass* — flagship §9%), so its BST closed form is inherently *coarser* the closer the mode sits to the boundary. The physical confinement-instability is the same fact: the top decays before it can hadronize (most boundary-localized $\rightarrow$ most unstable); charm/bottom are unstable hadrons; the light quarks are the ones whose ratios are sharp. One geometric quantity — boundary-distance — sets the mass, the precision, and the instability together.

Why the leptons are the control (the discriminant that makes it testable)

The leptons carry generation structure too, but they are colorless $\rightarrow$ boundary/free, not confined — their masses are clean pole masses, sharp at every generation. The quarks are colored $\rightarrow$ confined $\rightarrow$ interior/near-boundary limits — soft. So the split is *confinement*, not generation index, and the leptons are a built-in null control: if the blur were about “being generation 3,” the tau would blur; it doesn’t. That control is what turns a monotone observation into a falsifiable prediction.

The one dig — the discrete-series address (a,b) — and its three payoffs

The single open computation under all of this is the discrete-series address (a,b) of each fermion mode (F603's "sharply-pinned open core," flagged since June): which discrete (a,b) each quark generation sits at fixes its radial position, hence its boundary-distance. Computing the quark addresses delivers, from one object: 1. The generation count — the same discrete-series / radial-address structure the Gram-matrix radical uses (F713/F714); the addresses ARE the family labels. 2. The y_t / m_t Value — the radial overlap of the top (outermost address) with the spherical boundary condensate O is y_t (F603); its address decides whether the overlap saturates (the K964 m_t Value gap). 3. The precision-loss prediction — the boundary-distance read off the address is the input to the quantitative test below.

The test that would EARN it (quantitative — the anti-Cal-#27 guard)

Compute each quark's boundary-distance $d(a,b)$ from its discrete-series address, and check whether the relative precision of its BST mass form scales with d in a predicted way — not merely decreasing monotone. Concretely: - Does a single function $\epsilon(d)$ (the "width" of the near-boundary limit — e.g. the near-boundary behavior of the Harish-Chandra c-function / Gindikin Γ , which controls how sharply the overlap is defined) reproduce the observed ladder $0.09\% \rightarrow 0.5\% \rightarrow \text{coarse} \rightarrow 1.7\% \rightarrow 8\%$ at the computed d -values, within factors? - Cross-check (the same function, no new knobs): does $\epsilon(d)$ also track the *decay width / instability* (top width, charm/bottom lifetimes)? If the same boundary-distance predicts both the BST-form softness and the physical instability, the lens is real. - Null control: the leptons (colorless, sharp) must come out $\epsilon \approx 0$ (no blur) under the *same* $\epsilon(d)$ — otherwise the mechanism is confinement-specific and must say so. Promotes (Insight $\rightarrow$ Supported/Derived) only if $\epsilon(d)$ *predicts* the blur quantitatively; stays Insight if it merely tracks monotone (a monotone trend is not a derivation — Cal #27).

Discipline (held, because it's clean)

- Insight-tier: a reason to compute, not to bank. The convergence of Casey's intuition with the derived ordering (F603/K768) is a strong prior the *lens* is right — it is *not* evidence the precision-prediction holds. That distinction is the whole discipline here.
- Cal #27 fires hardest at the cleanest lead — a monotone trend that "feels" like a law is exactly where a fit hides. The guard is the quantitative $\epsilon(d)$ prediction + the same-function instability cross-check + the lepton null control.
- Shared machinery, not a new area: the dig is the discrete-series address, which the generation count already needs — so this is one computation, chartered alongside the count, not a second front.

Tiers / handoffs

- @Keeper — chartered at Insight-tier as directed; the earning test is the quantitative $\epsilon(d)$ -predicts-the-blur (+ instability cross-check + lepton null control), not the monotone trend. It shares the discrete-series-address dig with the generation count and the y_t/m_t Value — one dig, three payoffs — so I run it alongside the count, not as a separate lane. Nothing banked; convergence noted as a prior, not evidence.
- @Casey — your observation is a genuinely good lead, and here's why it's more than a pattern: the quarks lose sharpness up the generations but the *leptons don't*, which points the finger at confinement, not at "being generation three." The lens is one thing we already derived — how far a particle sits from the domain's edge sets its mass — read a new way: right at the edge, a mass isn't a clean number, it's a *limit* the particle approaches but never reaches, which is the same reason the top quark falls apart before it can settle down. So one distance sets the mass, the fuzziness, *and* the instability. But I'm holding it as a lead, not a result, until it *predicts* the fuzziness — the actual 0.09%-to-8% ladder — from the computed distances, not just moves the same direction. And the nice part: it's the same one dig (each quark's "address" in the geometry) that also gives us the generation count and the top-quark Yukawa — so I'm running all three off the one computation, which is your "one domain, one machine" again.
- @Elie / @Grace — if useful downstream: the quark discrete-series addresses (a,b) are the input; once I have them, the boundary-distances $d(a,b)$ feed the $\epsilon(d)$ test (blur) *and* cross-check against the measured

quark decay widths/lifetimes (instability) — same numbers, two checks.

Notes only; no toys/theorems claimed. F718 CHARTER (Insight-tier): quark generation = radial boundary-distance; it controls mass-ORDERING (DERIVED, F603/K768: mass = radial overlap with the boundary condensate; $t > c > u$ = most $\rightarrow$ least boundary-localized) AND mass-PRECISION (NEW lead: near-boundary = a *limit* not a pinned cell $\rightarrow$ soft/scheme-dependent observable $\rightarrow$ coarser form; = the confinement-instability). Leptons (colorless, sharp) = the null control $\rightarrow$ blur tracks confinement, not generation number. The ONE dig = the discrete-series address (a,b) (F603 June open core) $\rightarrow$ 3 payoffs: generation count (Gram radical), y_t/m_t radial overlap, precision-loss $\varepsilon(d)$. EARNING test: $\varepsilon(d)$ *predicts* the 0.09% $\rightarrow$ 8% ladder quantitatively (+ same-function instability cross-check + lepton null-control $\varepsilon \approx 0$), NOT monotone-by-eye. Discipline: convergence = a reason to compute not to bank; Cal #27 hardest at the cleanest lead. Runs alongside the count (one dig). — Lyra